Course: **Wireless Electromagnetic Technologies**

Final Project A.A. 2024-25

Combined Human-AI Design of a Patch Antenna Array

A.A. 2024-25

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Objective

The goal of this project is to design a printed antenna array for a real-world application using both traditional engineering methods and generative artificial intelligence (AI). Students will leverage ChatGPT to identify an innovative application, define technical requirements, and compare their manual design process with AI-driven suggestions.

Project Workflow

1. Starting Point: Reference Antenna

You will begin with a stacked patch antenna provided during the course. This antenna will serve as the foundation for your design.

2. Defining the Application Context

Use ChatGPT (or other LLM platform, Claude, GeminiAI,..) to explore application areas where an array of stacked patch might be useful. Suggested fields include (non only):
- Biomedical engineering (e.g., wearable health monitoring).
- Automotive (e.g., radar or communication systems).
- Aerospace (e.g., satellite communication).
- Security (e.g., perimeter monitoring).
- IoT and general wireless systems.

The patch must radiates toward the air.
Select one application and describe it in detail, including its purpose and relevance.

3. Defining Engineering Requirements

Use AI tool to help identify and justify the technical requirements for your chosen application. These requirements may include:
- Operating frequency range.
- Bandwidth.
- Beamwidth.
- Gain or directivity.
- Physical dimensions and constraints.

4. Designing the Antenna

4.1 Manual Design

Using the principles and formulas learned in class:
1. Design the single patch element to meet the defined requirements.
2. Extend the design to a linear or two-dimensional array configuration.
The ground plane must be considered, in any case, infinite as first-order approximation.

4.2 AI-Driven Design

Ask AI to suggest a design process for the same antenna array. Document the AI's approach and compare the design process and results with your manual calculations.

5. Simulation

Implement both your design and the AI-driven design in a simulation tool (Matlab MoM). Analyze and compare the performance metrics, such as:
- Return loss.
- Gain patterns
- Radiation efficiency
- Side lobes
- Bandwidth
- Array Coefficients (Tapering Efficiency)

6. Documentation and Comparison (<= 20 pages)

Write a report detailing:
1. The application context and its requirements
2. The design process (manual and AI-driven).
3. Simulation results and performance analysis.
4. A **critical comparison** of the manual design and the AI-driven design, highlighting:
- Similarities and differences.
- Strengths and weaknesses of each approach.
- Lessons learned about the use of AI in engineering.

Submission Requirements

1. Report: A structured document including:
- Title page.
- Table of contents.
- Main sections as outlined above.
- Clear diagrams and simulation results.
2. Simulation Files: Include all necessary files for verification.
3. Presentation: A brief 5-minute summary of your work, to be presented during the final session.

Evaluation Criteria

- Creativity (up to 6 units/30): How innovative is the application and the design approach?
- Technical Accuracy (9): Are the manual calculations and designs sound?
- Use of AI (5): How effectively was AI integrated into the workflow?
- Simulation Results (6): How well does the antenna perform based on the requirements?

- Documentation (4): Is the report clear, complete, and professional?

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Teams: 3 members
Deliverable: Technical Report <= 20pp (PDF Version). Including diagrams and explanations
Deadline: January 25.2025.
Rerference: Alessio Mostaccio
Project Validity: forever !

The delivery of the project within the indicated deadline is a pre-requirement for the final exams.

SUGGESTIONS

1. Do not report the full theory described in class, but only the used equations (with no intermediate manipulations).
2. The choices of the materials must be motivated. Refer to commercial substrates with documentable thicknesses (Manufacturer, code)
3. Identify the geometric parameters that can be optimized and their variation domain
4. Define a metric to evaluate how good an antenna configuration is. For example, the reflection coefficient module in the center band.
5. At the beginning of the work with MoM, perform some tests by varying the density of the mesh. Represent the metric chosen when the spacing varies (for example, the number of samples per wavelength). The optimal choice of gridding is that within the interval in which this graph is rather stable.
6. Represent the results obtained from intermediate attempts with MoM by the help of parametric diagrams. For example, if you decide to change two geometrical parameters p_1 and p_2 in the neighborhood of initial estimates, and you evaluate in various cases the reflection coefficient at the center frequency, then all the numerical attempts can be resumed by a contour plot $\Gamma(p_1, p_2)$. You can use the CONTOUR function of Matlab. A graph of this type allows understanding where the optimal solution is. I recommend starting from a coarse sampling of the parameters and then refine where it is needed.
7. Instead, if just a single parameter is changed, then the various solutions can be represented with a graph of the type $\Gamma(p_1)$
8. It is not useful to report only the final solution, while it is very appreciated to have described in a concise diagram the followed procedure. Consider using flowcharts.
9. All the diagrams shown in the report must be associated with a caption (ex**- Fig. 3**: *patch radiation diagram*) and must be referenced and commented in the text.
10. Pay attention to the decimal digits of the reported numbers. It makes no sense to provide so many decimal digits when there is no provision for an implementation process with the necessary accuracy. Stop at the tenth of a millimeter, e.g. x.yz cm or x.y mm

11. Page format suggestion: text Font: Arial 11 (not less than 10) or Times New Romans 12 (not less than 11) . Organize the report is numbered sections and subsection